ERDOS RESEARCH · POSITION PAPER

A New Framework for Forward Deployed Engineering
The Expertise-to-Execution Stack

> **A Forward Deployed Engineer transforms human expertise into organisational capability through successive layers of abstraction.** *At each layer the unit of abstraction changes, and each layer answers a different question.*

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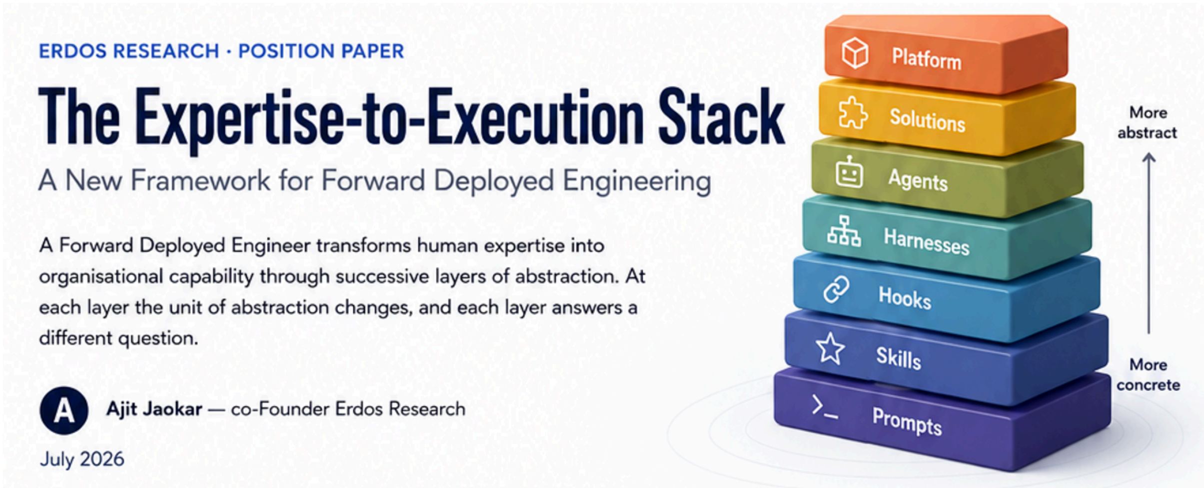
Abstract

The Forward Deployed Engineer (FDE) is usually defined by where they sit: an engineer embedded alongside the customer. This paper argues that location is the least important thing about the role. What an FDE actually does is convert tacit human expertise into durable organisational capability — and that conversion follows a predictable architecture, which we call the **Expertise-to-Execution Stack**. The stack has seven layers — Prompts, Skills, Hooks, Harnesses, Agents, Solutions, and Platform — through which knowledge is progressively operationalised. Two properties raise it above a mere taxonomy. First, the unit of abstraction changes at every layer: each layer makes a different kind of artefact reusable. Second, each layer answers a different question. Read together, these properties give the FDE role a definition that is both sharper and more ambitious than "an AI engineer who works with customers."



1. The FDE Role is more than a ‘seating plan’

Ask what a Forward Deployed Engineer is and the usual answer describes a seating plan. They are the engineer who leaves the product team, sits with the customer, and makes the software work in the customer’s messy reality. All true — and all beside the point. That description names the FDE’s *position*; it says nothing about the FDE’s *product*.

The trouble with a positional definition is that it invites positional metrics.

If the FDE is “the engineer near the customer,” success starts to look like tickets closed, integrations shipped, and demos delivered. These are real, but they are the residue of the work, not the work itself — and they evaporate. The integration is superseded, the demo forgotten, and the engineer moves to the next account carrying the hard-won knowledge in their head. Under this definition the most valuable thing an FDE produces — an understanding of how this domain actually works — is precisely the thing the organisation never captures.

This paper takes a different position. The FDE’s real output is not code and not proximity. It is **capability**: something the organisation can do reliably tomorrow that it could not do reliably yesterday, and can keep doing after the engineer has gone.

2. The position: transformation, not location

We propose the following definition.

> **A Forward Deployed Engineer transforms human expertise into organisational capability through successive layers of abstraction.**

The raw material is tacit knowledge — the judgement, heuristics, and hard-won context that currently live only in the heads of experts and in the muscle memory of teams. The finished product is capability the organisation owns: repeatable, inspectable, and independent of any single person. Between the two lies a sequence of transformations, each of which takes the output of the layer below and makes it reusable in a new way.

Tacit Knowledge → Prompts → Skills → Hooks → Harnesses → Agents → Solutions

On this view the FDE is not primarily writing software. They are performing a chain of abstraction: catching expertise the moment it is expressed, fixing it in a durable form, and then progressively composing those durable forms into systems. Code is one medium in which this happens, but the object being manufactured is organisational competence. That is a far more powerful account of the role than “an AI engineer working with customers.”

3. The Expertise-to-Execution Stack

The transformation is not a smooth ramp; it moves through discrete layers, and each layer composes the one beneath it — a higher layer is built out of the assets produced lower down. Seven layers make up the stack.

- **Prompts — one-off instruction.** The atom. A single instruction issued to a model to explore, experiment, or prototype: explain this architecture, summarise this document, draft these requirements. Prompts are deliberately disposable; their value is in the moment.
- **Skills — reusable expertise.** The first act of capture. When a prompt proves its worth, the FDE lifts it out of the conversation and turns it into a skill: a named, reusable unit encoding how a task should be done — a stakeholder interview, an architecture review, a risk assessment. The organisation stops reinventing the instruction each time.
- **Hooks — event-driven triggers.** A skill is inert until something calls it. A hook binds a skill to an event: a pull request opened, a document uploaded, a ticket created, an anomaly detected. Hooks move skills from “run when someone remembers” to “run when the world changes.”
- **Harnesses — reusable workflows.** Few outcomes come from a single skill. A harness chains skills into a repeatable process, usually with a human review step: interview → extract requirements → generate personas → prioritise → draft PRD → risk review. A harness is a business process expressed as composed skills.
- **Agents — owners of outcomes.** Above the harness sits something that decides. An agent owns a responsibility and chooses which skills to call, when, and in what order — drawing on memory, tools, hooks, and harnesses. The shift is from executing a process to owning a result.
- **Solutions — complete capabilities.** Agents, skills, hooks, and harnesses assemble into a coherent product that delivers a business capability end to end — an enterprise AI delivery platform, say, with its own dashboards, memory, and governance.
- **Platform — the organisational operating system.** At the top, the accumulated solutions, agents, and skills become shared infrastructure: a system through which the whole organisation builds and improves its AI capability. The platform is where individual capabilities compound into an institutional one.

Each layer is built from the layer below. That compositional property is what makes it a stack rather than a list — and it is what lets an organisation climb.

4. The first invariant: the unit of abstraction changes at every layer

It would be easy to read the stack as "the same thing, but bigger" — ever-larger bundles of automation. That reading misses the point. What actually changes as you climb is the *kind of thing being made reusable*.

Layer	Primary asset	Unit of reuse	Created by
Prompt	Instruction	Conversation	Human
Skill	Expertise	Individual	Practitioner
Hook	Event	Trigger	System
Harness	Process	Team	Organisation
Agent	Responsibility	Function	AI + Human
Solution	Capability	Business	Enterprise
Platform	Ecosystem	Organisation	Enterprise Architecture

Stated as a progression, each layer reuses something different:

- **Prompts reuse** language — a form of words that got a good result.
- **Skills reuse** expertise — the knowledge of how a task should be done.
- **Hooks reuse** events — the moments at which action should be taken.
- **Harnesses reuse** business processes — the way multiple steps fit together.
- **Agents reuse** responsibilities — the ownership of an outcome.
- **Solutions reuse** business capabilities — whole functions the organisation can perform.
- **Platforms reuse** organisational knowledge — the accumulated way the institution works.

This is the observation rarely stated explicitly, and it is the load-bearing idea of the framework. The stack is not a hierarchy of size but a hierarchy of *abstraction*: each layer takes something concrete from below and turns a new aspect of it into a durable, reusable asset. That is why the framework describes something deeper than "building AI applications." It describes how expertise is progressively operationalised — how something that begins as one person's judgement ends as an institution's capability.

5. The second invariant: every layer answers a different question

There is a companion property, and it gives the stack its conceptual spine. Each layer answers a distinct question about the work.

Layer	Question it answers
Prompt	What should the AI do right now?
Skill	How should this task always be done?
Hook	When should it happen?
Harness	How do multiple skills work together?
Agent	Who owns this responsibility?
Solution	What business capability does this provide?
Platform	How does the organisation continuously improve?

Read down the column and a natural progression appears: from *what* and *how*, through *when* and *how-together*, to *who*, and finally to *what-capability* and *how-to-improve*. The questions move from the immediate and tactical to the institutional and strategic.

Knowing which question you are answering tells you which layer you are working at — and, just as usefully, which layer you have not yet reached. A team drowning in one-off prompts has not answered "how should this always be done." An organisation with excellent skills but no hooks has not answered "when should this happen." The questions are a diagnostic as much as a description.

6. The FDE skills substrate

Redefining the role has a practical consequence: it tells us what an FDE must actually be good at. Because the work is capture-and-composition, the FDE's own competence is best described as a substrate of reusable skills across six capability areas — the skills from which every higher layer is built.

- **Discovery Skills.** Stakeholder Interview · Requirement Extraction · Business Problem Definition · Success Metrics · Current-State Assessment
- **Technical Skills.** Architecture Review · Integration Mapping · API Discovery · Data-Flow Mapping · Security Review
- **AI Skills.** Model Selection · Prompt Engineering · Evaluation · Agent Design · RAG Design · Tool Selection
- **Governance Skills.** Risk Assessment · Compliance Review · Human Oversight · AI Policy Mapping · Data Governance
- **Delivery Skills.** PRD Generation · Sprint Planning · Backlog Creation · Acceptance Criteria · Demo Planning
- **Communication Skills.** Executive Summary · Architecture Diagrams · Position Papers · Proposal Writing · Workshop Facilitation

These skills are the raw materials. The higher layers are simply this substrate, composed: hooks decide when a skill fires; harnesses chain skills into processes such as the Customer Discovery harness (interview → requirements → architecture → governance → PRD →

implementation plan); and agents — a Discovery Agent, a Solution Architect Agent, a Governance Agent — orchestrate whole responsibilities on top. Invest in the substrate and every layer above becomes cheaper to build.

7. From stack to maturity model

Because the layers compose, an organisation's position on the stack is a measure of its maturity. Read as stages, the same seven layers become a Capability Maturity Model for AI engineering — with an eighth rung for the organisation that turns its platform into a continuously improving operating system.

Level	Stage	What it represents
1	Prompts	One-off work
2	Skills	Reusable expertise
3	Hooks	Event-driven execution
4	Harnesses	Reusable business workflows
5	Agents	Autonomous execution
6	Solutions	Enterprise AI products
7	Platform	An organisational operating system

Most organisations today live at Levels 1–2 — plenty of prompting, a handful of reusable skills — and mistake activity for capability. Maturity is not measured by how much AI an organisation uses, but by how high up the stack its reusable assets reach. The model gives leaders somewhere to stand and a direction of travel: the next rung is always the next question they cannot yet answer.

8. Implications

For the role

FDEs should be hired and measured for their ability to move expertise up the stack, not for tickets closed. The best FDE is not the one who ships the most; it is the one who leaves behind assets that keep working — skills, harnesses, and agents that outlive the engagement. Measured this way, the role stops looking like senior support and starts looking like capability manufacturing.

For the organisation

The stack is an argument for treating expertise as infrastructure. Capture at the skill layer is the single highest-leverage habit an organisation can build, because everything above it composes from there. An organisation that never rises above prompting is paying, again and again, to re-derive knowledge it already had.

For education and curriculum

The progression is also a syllabus. Each layer is a discipline in its own right — prompt engineering, skill engineering, hook engineering, harness engineering, agent engineering, enterprise solutions, and platform engineering — and each builds on the last. A curriculum organised around the stack teaches not merely how to use AI, but how to build increasingly sophisticated capability systematically. It is a natural spine for the Erdos programme: every course a rung, every rung a new question.

For domain experts

This idea can be expanded to domain experts - making FDE an ideal role for Domain experts through this methodology

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9. Position

The Forward Deployed Engineer is not "an AI engineer who works with customers." They are the person who transforms what an organisation knows into what it can reliably do — layer by layer, question by question. Define the role that way and the rest follows: how to hire for it, how to measure it, how to teach it. The Expertise-to-Execution Stack is the map of that transformation, and its two invariants — a changing unit of abstraction and a changing question at every layer — are what make it more than a list of tools. They describe how human expertise becomes organisational capability.

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